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Enhancement of Color Images By Efficient Demosaicing

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Abstract

We propose an efficient method for reconstructing a full-color image from its partially sampled version. The suggested algorithm is non-iterative and is based on the properties of the human visual system. While most state-of-the-art algorithms invest a great deal of computational effort in the enhancement of the reconstructed image to overcome the color artifacts, we focus on eliminating most of the color artifacts in the initial stages of the algorithm. This approach enables us to correct a specific kind of color artifact, caused by the aliasing involved in the sampling process. Our correction method is based on the relatively low sensitivity of the human eye to high transitions of the hue component of a color image. We tested our algorithm on several problematic images and found it to often be significantly superior to state-of-the-art algorithms, without consuming high computational power.

Index Terms—Demosaicing, interpolation, color spaces.

1 Introduction

It is well known that in order to display a digital color image, one needs to specify only three values for each pixel, i.e. the red, green, and blue values (often called the color channels) at the corresponding location. Due to technology limitations, however, most charge-coupled-device (CCD) cameras provide a single value for each pixel, that is, either red, green, or blue. The resulting image is often called a "mosaic". The problem of demosaicing is to reconstruct the original, full-color, image from its subsampled version, and is usually regarded as an interpolation problem.

Many algorithms have been proposed over the last decade for the demosaicing problem; see, e.g. [1], [2], [3], [4], [5], [6], [7] to mention a few. These algorithms can be broadly divided into three categories. In the first category, standard interpolation techniques, such as nearest neighbor, bilinear, or cubic interpolation are used to interpolate the missing pixels. Such methods are computationally efficient, but result in severe color artifacts. The algorithms belonging to this category completely ignore the color attributes of the image in the interpolation process, explaining their degraded visual quality.

The second category of algorithms takes into account the cross-correlation among the three color channels. Common use of this correlation is to assume that the hue changes smoothly between

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